使用 Python

ONTAP Automation

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使用 Python

Python 是一种受欢迎的编程语言，尤其适用于数据中心自动化。您可以使用 Python 的原生功能或通过 ONTAP Python 客户端库访问 NetApp REST API 。

NetApp ONTAP Python 客户端库

从 ONTAP 9.6 开始，您可以选择使用 Python 客户端库访问 ONTAP REST API 。该库可为客户端开发环境提供多种基础服务，例如连接管理，异步处理，异常处理和错误消息。下载并安装 ONTAP Python 客户端库软件包后，您可以快速创建强大的 Python 脚本，以支持 ONTAP 部署的自动化。

> NetApp 维护一个 GitHub 存储库，其中包含代码示例和其他有用信息。您可以导航到 *examples* 文件夹以使用 Python 客户端库访问示例。

"ONTAP REST Python GitHub 存储库"

正在准备使用 Python 客户端库

在使用 Python 客户端库之前，您应准备好环境。

基本要求

您必须使用 Python 3.5 或更高版本。此外，还需要以下软件包：

- 请求 2.22.1 或更高版本
- requests-toolbelt 0.9.1 或更高版本
- marshmallow 3.2.1 或更高版本

软件包名称和版本

Python 客户端库软件包的名称是 * 。 netapp-ontap* 。与软件包关联的版本是生成库时使用的 ONTAP 主要版本号和次要版本号的组合，以及 ONTAP 版本中客户端的次要版本。例如，有效版本号包括： 9.2.1 ， 9.6.2 和 9.1.1 。

安装

您必须使用 pip 从 Python Package Index （ Pypi ）网站安装 NetApp_ONTAP 软件包。

文档和其他资源

有关文档和其他资源的链接，请参见 _NetApp 开发人员网络 _ 。

"PyPI： NetApp ONTAP 9.10.1"

"适用于 9.10.1 的 NetApp PCL-文档"

"NetApp 开发人员网络： ONTAP RESTful API"

用于检索集群配置的脚本

以下脚本提供了一个简单的示例，用于说明如何使用 Python 客户端库。您可以在命令行界面中使用 Python 3 运行此脚本，以检索 ONTAP 集群配置。

```
##-----------------------------------------------------------------------
#
# Description: Python script to retrieve the cluster configuration.
#
# Usage example:
#
# python3 get_cluster.py
#
#
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#
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# that the software application product is distributed pursuant to terms
# no less restrictive than those set forth herein.
#
##-----------------------------------------------------------------------
# Global configuration for the library
from netapp_ontap import config
# Support for the connection to ONTAP
from netapp_ontap import HostConnection
# Specific API needed for this script
from netapp_ontap.resources import Cluster
# Create connection to the ONTAP management LIF
conn = HostConnection("10.236.252.97", username="admin",
password="mypassword", verify=False)
# Set connection as the default for all API calls
config.CONNECTION = conn
# Create new cluster object
clus = Cluster()
# Issue REST API call
clus.get()
# Display the cluster configuration
print(clus)
....
```

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